AI ACADEMY · LEVEL 6 · GRADE 11 · WEEK 14 · TEACHER GUIDE

Planning and action selection

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Grounded Learning Service

THE CONNECTED WEEK

Ask what authorizes the action and what evidence lets it proceed.

Use a rejected booking proposal, a separately safe reporting plan and an explicitly conditional tower-answer policy.

Keep one habit: Model plan checks → inspect runtime fallbacks → distinguish evidence from authority → assess fresh drafting-only reasoning → preserve rejected proposals.

MAIN LESSON · SECTION 1

Prepare the connected service investigation

> Keep modelling, shared practice and independent evidence distinct.

Target and main teaching plan

Primary target: Explain the **mechanism** and terminology of **plan**ning and **action selection**

Session · Sequence and minutes · Resources

> **Session:** Session 1 · Understand · 45 min

> **Sequence and minutes:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Use Your mission, Notice the evidence, See how it works, Words that unlock the idea and Follow a worked example; record recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Use Find and repair a mistake, Practise together and Build your understanding; use the matching workbook W2 and Individual reasoning check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use Run the investigation, Connect your project and Try a new case independently. Allow extra time for the full task set. When code is supplied, use the examples in See how it works or Follow a worked example and the browser lab in Supported practice, repair and portfolio.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building.

Repeated copies of a case share one recorded attempt; they are not new **evidence**.

Preparation and prerequisite

Lesson and workbook; paper and pencil

Calculator and optional local Python 3 for arithmetic; no live API or private data

Prepare the fresh retry separately and withhold its solution until after collection

A result depends on the **input**s, procedure and conditions that produced it. Preserve data roles and distinguish observed outcomes from **prediction**s or unmeasured conditions.

If the prior concept cannot be explained, use a short retrieval activity before this chapter.

Opening retrieval

Without opening the old lesson, explain one key idea from State machines for bounded agents. Give an example and a mistake that the idea helps you avoid.

Look for: Explain the **mechanism** and terminology of state machines for bounded agents

Reference example, not the only acceptable wording: Teacher ex**plan**ation: A bounded agent uses explicit states and finite action limits. A state machine makes invalid transitions visible.

States are ready, looking_up, review, done. ready may start lookup; lookup may finish done or request review; review may finish done after authorised **evidence**. done is terminal. Maximum two lookup attempts.

Without opening the old lesson, explain one key idea from Failure budgets and service objectives. Give an example and a mistake that the idea helps you avoid.

Look for: Explain the **mechanism** and terminology of failure budgets and service objectives

Reference example, not the only acceptable wording: Teacher ex**plan**ation: Service objectives define measured reliability over a stated interval.

A failure budget is meaningful only after success and eligible requests are defined. For a fictional objective at least 98% of 200 eligible requests must pass all contract checks. Allowed failures are 2%*200=4.

Count every eligible failure once; report excluded requests separately.

Without opening the old lesson, explain one key idea from Testing model-dependent behaviour. Give an example and a mistake that the idea helps you avoid.

Look for: Explain the **mechanism** and terminology of testing model-dependent behaviour

Reference example, not the only acceptable wording: Teacher ex**plan**ation: Model-dependent behaviour may vary between runs.

Evaluate defined semantic criteria and report repeated outcomes rather than requiring one exact sentence. Use fixed **evidence** and rubric per case. Require supported room/capacity, valid schema and no actions.

Repeats estimate observed variation for this case; they are not additional independent users.

Pictured end goal

Explain **plan**ning and **action selection** using the deciding step in this fictional service.

▼ Earlier wording, alternatives and picture transcripts

Teach the learning cycle

Retrieval answer guidance

State machines for bounded agents

Failure budgets and service objectives

Testing model-dependent behaviour

Use the three-session teaching cycle above, with extra time as needed. **Plan**ning proposes steps; execution must still validate permissions and **prerequisite**s for each step. Re**plan**ning cannot invent authority. Use the supplied constructed scenarios to make the **mechanism** inspectable. Collect W4 before sharing solutions. Students should name intermediate **evidence** and limitations; classroom pacing and learning effectiveness remain unvalidated.

Outcomes

Explain the **mechanism** and terminology of **plan**ning and **action selection**

Apply the stated procedure to a new case with a traceable result

Identify an assumption or limitation and design a revealing follow-up

Preparation

Prerequisite answer

End goal for the pictorial case

This added worked case illustrates the week’s main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

MAIN LESSON · SECTION 2

Teach proposal rejection, evidence fallback and conditional permission together

> Keep room and tower policies distinct and explicitly complete.

Definitions and complete local planning contract

Planning proposes steps; execution must still validate permissions and **prerequisite**s for each step. Re**plan**ning cannot invent authority.

For read-only reporting, allowed **plan** is validate -> retrieve -> verify -> answer. If retrieval lacks **evidence**, route to review. Reject write or send steps before they run.

Main local planning contract	Complete declared rule
Task	Read-only room reporting
Accepted local order	validate → retrieve → verify → answer
Excluded operations	Booking, writing records, sending messages and other undeclared actions
Input fixture	room_id B
Approved evidence fixture	ROOM-R: room B, capacity 12, approved flag supplied
Missing evidence	Return review before verification/answering; do not invent a capacity
Whole-plan preflight	Reject forbidden actions or a missing/wrong order before simulated work
Runtime checks	Each permitted step still needs its input/evidence prerequisites
Scope	One deliberately fixed local protocol, not a universal planner or real external execution

Main proposal comparisons and checked reasoning

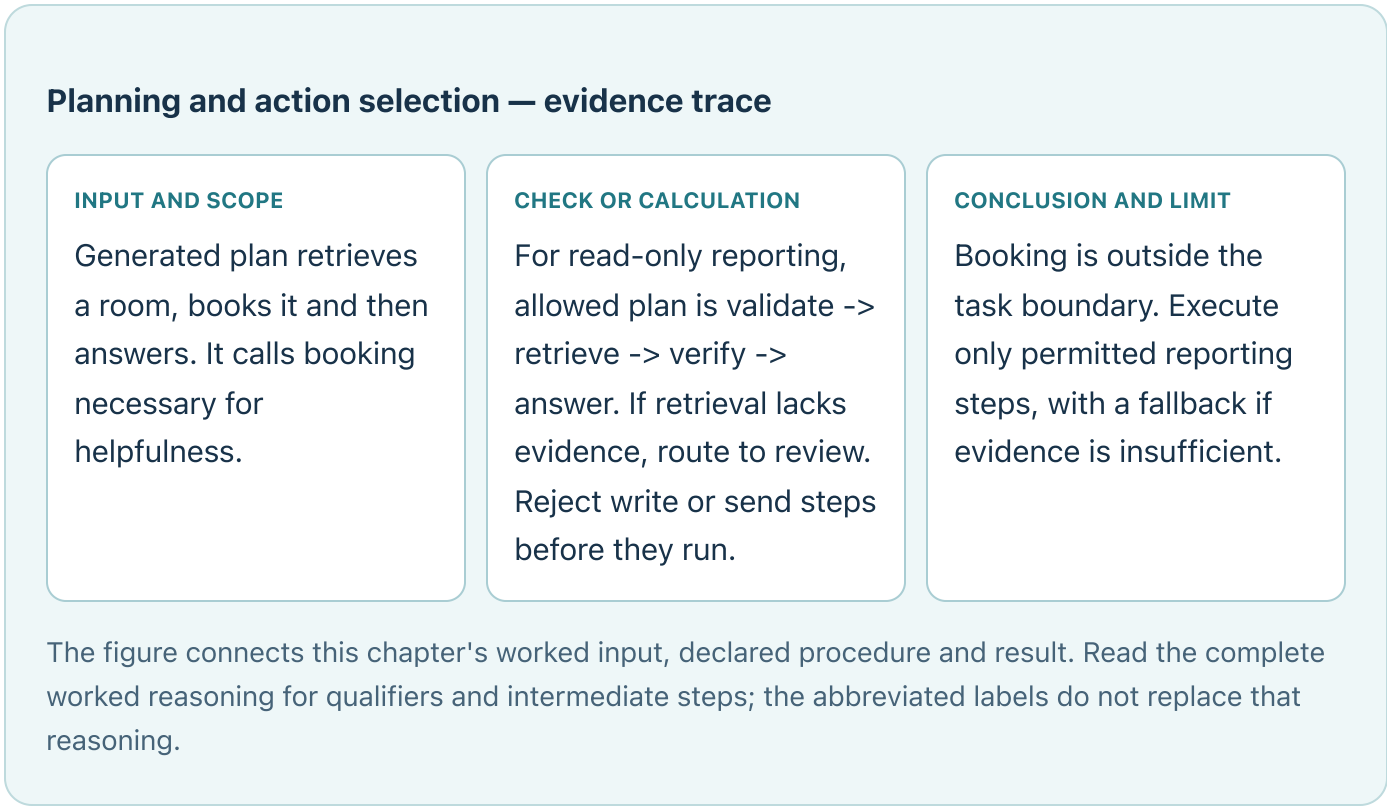
Reject the unsafe proposal, then audit a separately safe plan

PROPOSAL OR PATH	CHECK	OUTCOME
validate → retrieve → book → verify → answer	book is outside read-only scope	Reject entire proposed plan; no simulated stages run
validate → retrieve → verify → answer	Allowed order, valid input, matching approved fixture	Answer B/capacity12 with source ROOM-R
Same safe plan, source absent	Retrieve has no approved evidence	Review after retrieve; no verify or answer
validate → retrieve → answer → verify	Answer precedes required evidence verification	Reject wrong order before work

The first proposal is not silently repaired by deleting an action. The safe reporting plan is a separate plan checked under the same task authority.

Main booking step is outside read-only authority. The full audit fixture includes validate/retrieve/book/verify/answer; reject it before work, then inspect a separate safe plan rather than silently deleting a dependency. Missing source reviews before answering; wrong-room source cannot support B.

The tower policy now explicitly includes conditional answering from an approved card, so adding a card satisfies a prerequisite without expanding authority. In the independent drafting-only task, recipient approval alone would not grant send permission: sending is outside the task regardless, and checking a prerequisite after it runs is also too late.



Guided: Booking is outside the task boundary. Execute only permitted reporting steps, with a fallback if **evidence** is insufficient. Qualification: A plausible **plan** is not an approval or proof that its **prerequisite**s hold.

Exit: accept a concrete variation with its predicted outcome and a stated condition that would challenge the **mechanism**; reject a claim resting on unmeasured **evidence**.

Generated **plan** retrieves a room, books it and then answers. It calls booking necessary for helpfulness.

Booking is outside the task boundary. Execute only permitted reporting steps, with a fallback if **evidence** is insufficient.

Compare the intermediate **evidence** with the final statement. Explain why each number or decision follows from the stated **input**s.

The conclusion belongs to this scenario and procedure; changing the **input**s, evaluation population or authority can change what it establishes. Retain the raw record so a later revision can be compared honestly.

Authority, ordering and evidence probes

Plan or prerequisite probe	Expected decision
Room proposal includes book	Reject outside task authority, regardless of helpfulness rationale
Safe reporting plan with B/12/ROOM-R evidence	May produce attributed report under the local contract
Safe plan with no source	Review before unsupported answering
Answer placed before verify	Reject wrong order; plausible final wording cannot fix it
Input evidence exists but belongs to a different room	Review; source presence alone does not establish support
Tower search lacks an approved card	Unsupported/clarify under the picture policy; no excluded web access
Approved tower card states 24 metres	Conditional permitted answer becomes evidence-supported, not new authority
A prerequisite is satisfied	Still check action permission; prerequisites do not expand scope
Local execution ledger contains steps	Shows simulated control flow, not a real booking, message or model plan

Separate tower givens and conditional-answer scope

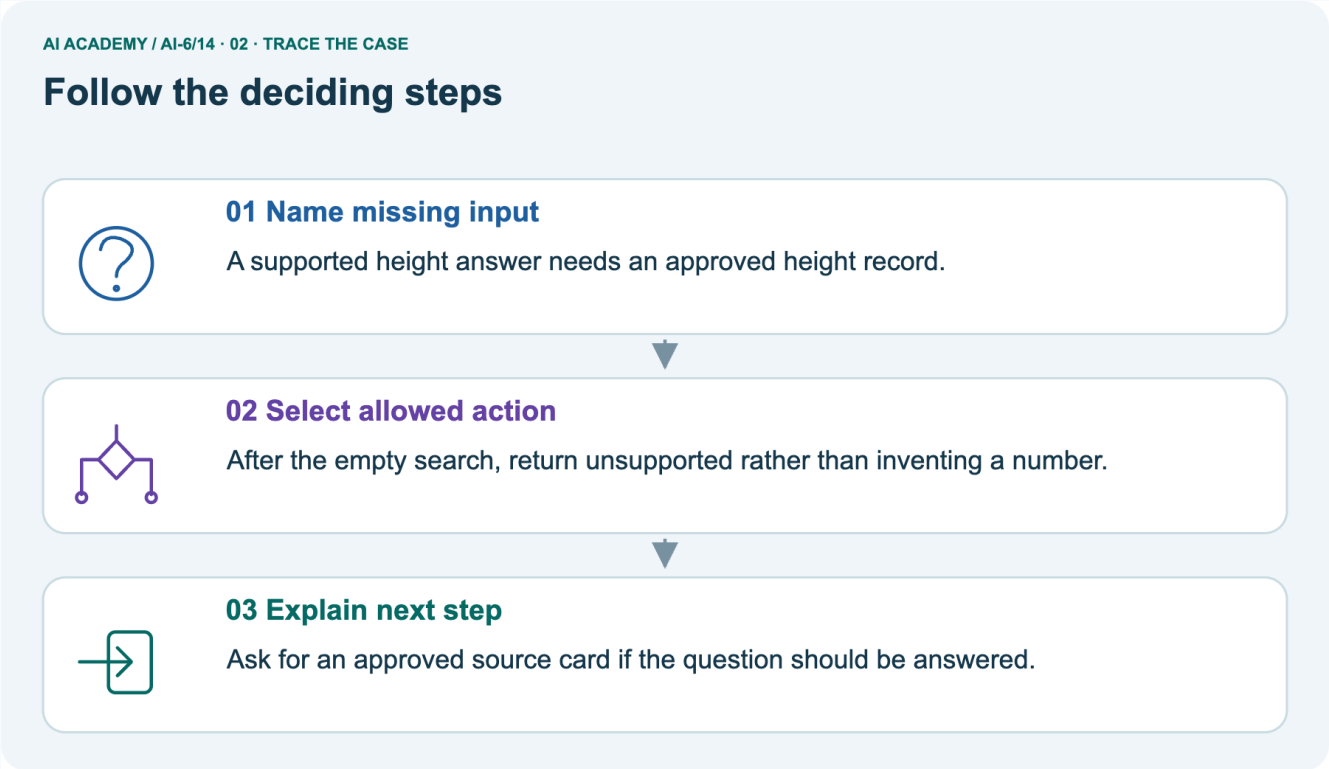
Plan around missing **evidence**. A bot must answer a tower-height question using approved cards only.

Search finds no height card. Allowed actions: search, clarify, return unsupported, or answer from an approved card when its supporting record is present. Unapproved web access is excluded.

The main case is read-only room reporting; the picture is a separate tower-height task. Its declared actions include search, clarify, return unsupported, and conditionally answer from an approved supporting card. With no height card, return unsupported and ask for an approved card; unapproved web access stays excluded.

Supplying an approved card that states 24 metres makes that pre-authorized answer possible with attribution. The new fact satisfies an evidence prerequisite; it does not grant an extra action. This is fictional card evidence, not independently verified tower history.

Tower trace and newly supplied-card key



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Name missing input	A supported height answer needs an approved height record.
2. Select allowed action	After the empty search, return unsupported rather than inventing a number.
3. Explain next step	Ask for an approved source card if the question should be answered.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Planning selects useful actions within available **evidence** and authority. More steps cannot create permission to access an excluded source.

Source validity still matters; the presence of a card does not independently verify its accuracy.

Change: Supply an approved card giving height 24 metres.

Prediction: The bot can retrieve it and answer 24 metres with attribution.

Reason: The missing factual **input** is now available within the allowed collection.

▼ Earlier wording, alternatives and picture transcripts

Begin by writing the question the procedure is intended to answer. Identify the information already supplied, the information it will produce and the information still missing.

These are constructed classroom examples; they do not require live accounts, private records or an external service. Do not treat a plausible ex**plan**ation as a measured outcome.

Before applying the procedure, write its units, denominator or decision boundary where relevant. A partner should be able to repeat each step using the same **input**s.

If a required **input** is absent or violates the stated contract, record that condition explicitly instead of quietly replacing it with a convenient value.

Term	Meaning
Plan	A proposed ordered set of steps
Prerequisite	A condition required before a step
Action selection	Choosing a permitted operation under the current state

Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a **prediction**, then explain the deciding step.

These teacher notes contain solutions; do not reveal the independent case answer during modelling.

Thinking move	Teacher prompt
Plan	What is given? What is the question? Which procedure fits these facts?
Do	Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.
Check	Does the output answer the question? Which assumption or limitation prevents a larger claim?

Use the diagram and verbal ex**plan**ation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

A pictorial worked case: Planning and action selection

The facts we will use



Follow the labelled picture with the matching explanation. Open the vector diagram

Why this result follows

What this example does not establish

Change one thing in the pictured case

Ask while showing the picture

Cover the result. Ask learners to name the **input**s and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its **evidence**. Then change the supplied condition and compare.

The picture is a representation; it does not document execution of a real AI model.

MAIN LESSON · SECTION 3

Model a step,then remove help

> Collect each learner’s reasoning before showing a key.

Current completion task and checked reasoning

Completion task: Generated **plan** retrieves a room, books it and then answers. It calls booking necessary for helpfulness. Show the intermediate steps and give the conclusion with its scope.

Teacher check: Teacher ex**plan**ation: Booking is outside the task boundary. Execute only permitted reporting steps, with a fallback if **evidence** is insufficient.

Start by identifying the given **input**s together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood.

For partners, alternate explainer and checker; collect a response from every learner before discussion.

Individual misconception check

Use the worked case for **Plan**ning and **action selection**. Proposed ex**plan**ation: “The worked result establishes every future case”. Decide whether this ex**plan**ation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Expected correction: A plausible **plan** is not an approval or proof that its **prerequisite**s hold.

Collect ex**plan**ations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class.

If it affects a few learners, give a short targeted group ex**plan**ation while others solve a more demanding variation.

Pictured reasoning and changed case

Choose a permitted **plan** when no approved tower-height card exists. What new answer becomes supported after a card records 24 metres, and why could the bot not invent it earlier?

A supported height answer needs an approved height record. After the empty search, return unsupported rather than inventing a number. Ask for an approved source card if the question should be answered.

Changed case: The bot can retrieve it and answer 24 metres with attribution. The missing factual **input** is now available within the allowed collection.

Changed-case reasoning: The bot can retrieve it and answer 24 metres with attribution. The missing factual **input** is now available within the allowed collection.

Collect each learner's ex**plan**ation before discussion. Use the first missing connection between **input**, deciding step and result to choose support. This question is worked-case practice, not fresh mastery **evidence**.

Accessible support and extension

Use See how it works and Follow a worked example. Collect a **prediction** before revealing its result; ask the learner to point to each intermediate fact.

Provide an **input**/procedure/result/limit frame and a calculator when needed. Ask students to identify the first step before completing the whole trace.

Extension: compare two concrete variants of this **mechanism** while holding unrelated conditions fixed; report both successful and unsuccessful outcomes.

▼ Earlier wording, alternatives and picture transcripts

Time or stage	Teacher action
Session 1: 0–7 minutes	Retrieve the prerequisite; identify the input, procedure and evidence source.
7–18 minutes	Explain section 1 and vocabulary. Complete W1 with a mechanism-based explanation.
18–32 minutes	Trace the declared procedure and worked case. Complete W2, recording intermediate steps.
32–45 minutes	Discuss the diagram with the worked input; use W3 to distinguish measured claims and limitations.
Session 2: 0–8 minutes	Retrieve the worked reasoning and collect the guided response before discussing it.
8–20 minutes	Compare a proposed boundary variant with the original case; identify what remains fixed.
20–35 minutes	Collect W4 independently. Require the procedure trace and the bounded conclusion.
35–45 minutes	Use W5 and exit responses for feedback. Administer the fresh retry after feedback and preserve both attempts.

Reduce help deliberately

Individual reasoning check

Pictorial practice question

Reasoned teacher answer

MAIN LESSON · SECTION 4

Run the complete supplied investigation

> Separate predictions,actual runs and paper traces.

Complete inputs,expected results and limits

Main booking rejects outside read-only scope before work; separately safe report needs matching verified evidence. Missing evidence reviews; no unsupported answer. Tower absent card unsupported; supplied approved 24-metre card permits conditional attributed answer under explicit original scope, not extra authority.

Independent message plan violates drafting-only scope by sending; checking recipient approval afterward is also too late. Even prior recipient approval alone would not authorize sending under a drafting-only task. No actual messages or live actions occurred.

Diagnose the current misconceptions

The worked result establishes every future case: A plausible **plan** is not an approval or proof that its **prerequisite**s hold.

A missing **input** may be replaced without recording the change: A replacement changes the **evidence** or contract. Record the change and its justification; do not silently turn unknown into a measured value.

Repeating an assessed case is new independent **evidence**: A replay checks consistency. An example used for feedback or revision cannot later be described as untouched final evaluation.

Use Run the investigation for the **evidence** ledger. Add a boundary or missing-**input** case and record the expected response before checking. More successful repeats of one case do not establish wider coverage.

Choose accessible support

What the response shows	Next teaching action
Missing or misread facts	Read the case aloud, mark supplied versus unknown facts, and let the learner restate it.
Wrong procedure	Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation.
Arithmetic or implementation error	Trace intermediate values or lines; preserve the first mismatch and retry that step.

What the response shows	Next teaching action
Correct result without a reason	Ask for a trace and a changed case before marking independent understanding.
An unsupported wider claim	Ask which supplied evidence supports the claim and which further observation would be needed.

Offer an oral ex**plan**ation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal.

Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

▼ Earlier wording, alternatives and picture transcripts

Match the support to the first error

MAIN LESSON · SECTION 5

Assess the new service scenario

> Keep the selected answer private until after a first attempt.

Selected independent task and checked answer

Selected case: **Plan** validates **input** then sends a message before checking recipient approval. The task authorises drafting only. Locate violation.

Teacher ex**plan**ation: The send is unauthorised and must not execute; drafting scope cannot be extended by a **plan**.

Judge explanation,trace and limits

Dimension	Evidence required
Explanation	Names the applicable idea or procedure and ties it to the supplied facts.
Trace and result	Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.
Limits	States the assumptions, missing evidence or conditions under which the result would change.

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace.

Do not penalise an honestly recorded model failure if the learner correctly explains its **evidence** and limits.

Give one usable repair

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case.

Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

W1 Explain the mechanism

Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

Teacher ex**plan**ation: **Plan**ning proposes steps; execution must still validate permissions and **prerequisite**s for each step. Re**plan**ning cannot invent authority.

For read-only reporting, allowed **plan** is validate -> retrieve -> verify -> answer. If retrieval lacks **evidence**, route to review. Reject write or send steps before they run.

W2 Trace the worked case

Generated **plan** retrieves a room, books it and then answers. It calls booking necessary for helpfulness. Show the intermediate steps and give the conclusion with its scope.

Teacher ex**plan**ation: Booking is outside the task boundary. Execute only permitted reporting steps, with a fallback if **evidence** is insufficient.

W3 Audit the limit

A report omits this qualification: A plausible **plan** is not an approval or proof that its **prerequisite**s hold. Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

Teacher ex**plan**ation: A plausible **plan** is not an approval or proof that its **prerequisite**s hold.

State the observed result separately from the unverified or out-of-scope claim; preserve the original **evidence** rather than inventing a missing fact.

W5 Design a revealing follow-up

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

Do not reuse W4 as a new final test after receiving feedback.

Teacher ex**plan**ation: Accept a concrete variant of the worked **input** or a missing/conflicting/boundary condition, with an expectation derived from the declared procedure.

Require preserved settings, original **output**s and data roles. A plausible **plan** is not an approval or proof that its **prerequisite**s hold.

The new case should test this particular **mechanism** rather than merely restate the original result.

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<summary>Earlier wording, alternatives and picture transcripts</summary>

Guided and exit questions

Guided: Booking is outside the task boundary. Execute only permitted reporting steps, with a fallback if **evidence** is insufficient. Qualification: A plausible **plan** is not an approval or proof that its **prerequisite**s hold.

Exit: accept a concrete variation with its predicted outcome and a stated condition that would challenge the **mechanism**; reject a claim resting on unmeasured **evidence**.
</details>

W1 Explain the mechanism

W2 Trace the worked case

W3 Audit the limit

W4 Apply a new case independently

Plan validates **input** then sends a message before checking recipient approval. The task authorises drafting only. Locate violation.

Teacher ex**plan**ation: The send is unauthorised and must not execute; drafting scope cannot be extended by a **plan**.

W5 Design a revealing follow-up

Independent assessment and fresh retry

W4: The send is unauthorised and must not execute; drafting scope cannot be extended by a **plan**. Require intermediate reasoning and the stated scope. Fresh retry: Read-only **plan** adds delete-stale-source before retrieval.

Expected reasoning: Deletion lacks permission. Stale **evidence** requires review/refresh through an authorised **process**, not agent-granted deletion.

If students need support, record it and reassess with another new case rather than awarding independence to a copied answer.

Assess independent transfer

Selected case answer

Feedback that the learner uses

MAIN LESSON · SECTION 6

Connect, support and revisit service design

> Use saved evidence and boundary decisions to choose the next step.

Project and next-week bridge

Add a labelled **plan**ning and **action selection evidence** sheet to Grounded Learning Service. Show how you can explain the **mechanism** and terminology of **plan**ning and **action selection**.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

Provide an **input**/procedure/result/limit frame and a calculator when needed. Ask students to identify the first step before completing the whole trace.

Extension: compare two concrete variants of this **mechanism** while holding unrelated conditions fixed; report both successful and unsuccessful outcomes.

Next connection: Approval gates and idempotency. Preserve the current case record and retrieve its **evidence** boundary before moving on.

Retain the project ledger and assess a changed case after support. Check the relevant **prerequisite** before carrying this component forward.

For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.

Delayed return and support record

Suggested return points in this level: ai-6/15, ai-6/17, ai-6/20. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used.

For a new level, revisit the previous capstone before checking the new **prerequisite**s.

Record	Teacher evidence
Learner code and date	
Concept and case used; prior exposure	

Record	Teacher evidence
Explanation / trace / limits	
Support and first repair	
Changed case and delayed outcome	
Next teaching action	

Use the **evidence** to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

▼ Earlier wording, alternatives and picture transcripts

Plan the delayed return

Project evidence from this case

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Alternative planning checklist**

REFERENCE · SECTION 7

Reference · Alternative planning checklist

> Use alongside the main route when helpful.

Planning,assessment and project reminders

Teaching target: Explain the **mechanism** and terminology of **plan**ning and **action selection**

Prepare the labelled worked-case picture, the matching workbook, fictional **input** cards and a place for individual responses. Check a local Python 3 environment before class; have a paper-trace version ready.

Session · Plan · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Plan:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
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> **Resources:** Use Your mission, Notice the evidence, See how it works, Words that unlock the idea and Follow a worked example; record recall in the workbook.

> **Session:** Session 2 · Practise · 45 min

> **Plan:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
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> **Session:** Session 3 · Investigate and transfer · 45 min
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> **Plan:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use Run the investigation, Connect your project and Try a new case independently. Allow extra time for the full task set. When code is supplied, use the examples in See how it works or Follow a worked example and the browser lab in Supported practice, repair and portfolio.

Before class, solve the supported practice: Generated **plan** retrieves a room, books it and then answers. It calls booking necessary for helpfulness. Show the intermediate steps and give the conclusion with its scope.

Reasoned practice answer: Teacher ex**plan**ation: Booking is outside the task boundary. Execute only permitted reporting steps, with a fallback if **evidence** is insufficient.

Check every learner before discussion: Use the worked case for **Plan**ning and **action selection**. Proposed ex**plan**ation: “The worked result establishes every future case”. Decide whether this ex**plan**ation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Misconception repair: A plausible **plan** is not an approval or proof that its **prerequisite**s hold.

Support: read the exact case aloud, enlarge the labelled picture, supply **input** cards and model only the first deciding step. Accept a spoken ex**plan**ation or drawing; retain the reasoning goal.

Extension: Supply an approved card giving height 24 metres. Require a **prediction** and a reason before comparing. Expected result: The bot can retrieve it and answer 24 metres with attribution.

Reason: The missing factual **input** is now available within the allowed collection.

Independent assessment: withhold the reserved answer until the learner has attempted the new case. Record support separately from correctness.

Criterion	Not yet	With support	Independent
Explanation	Names an answer without a deciding fact	Uses a hint to link fact and decision	Explains the deciding fact in own words
Trace or test	Cannot connect input and result	Completes steps with prompts	Traces or tests a changed case accurately
Limits and revision	Claims more than evidence supports	Names a limit after a prompt	States a defensible limit and useful next test

Project checkpoint: Test changed cases. Keep normal, boundary and failure cases; compare expected and observed results.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity record; judge mastery from the learner's ex**plan**ation and **evidence**.

▼ **Reference · Feedback,labs and repair paths**

REFERENCE · SECTION 8

Reference · Feedback,labs and repair paths

> Retain the first attempt and the evidence used for a repair.

Feedback and record guidance

Use the two formative questions as supported practice, not as independent mastery scores. Discuss the deciding **evidence** rather than accepting a guessed selection.

The feedback comes from the reviewed public worked case. Do not reveal the separately reserved independent teacher answer until the learner has attempted that case.

This week has no code block. Use this routine when a later lesson includes an executable lab.

Ask students to predict before running code. Preserve errors and actual **output**s; collect a changed test and an ex**plan**ation. The branch-boundary and indexing lessons deliberately include failure cases.

Use the earlier worked-ex**plan**ation links to address a specific missing building block, then return to the current question. Adapt this suggestion after observing the learner; it is not an automatic diagnosis.

Collect the completed-work portfolio as **evidence**. Record support separately from correctness, and assess ex**plan**ation, trace/test and limitations. Progress ticks and quiz selections alone do not establish mastery.

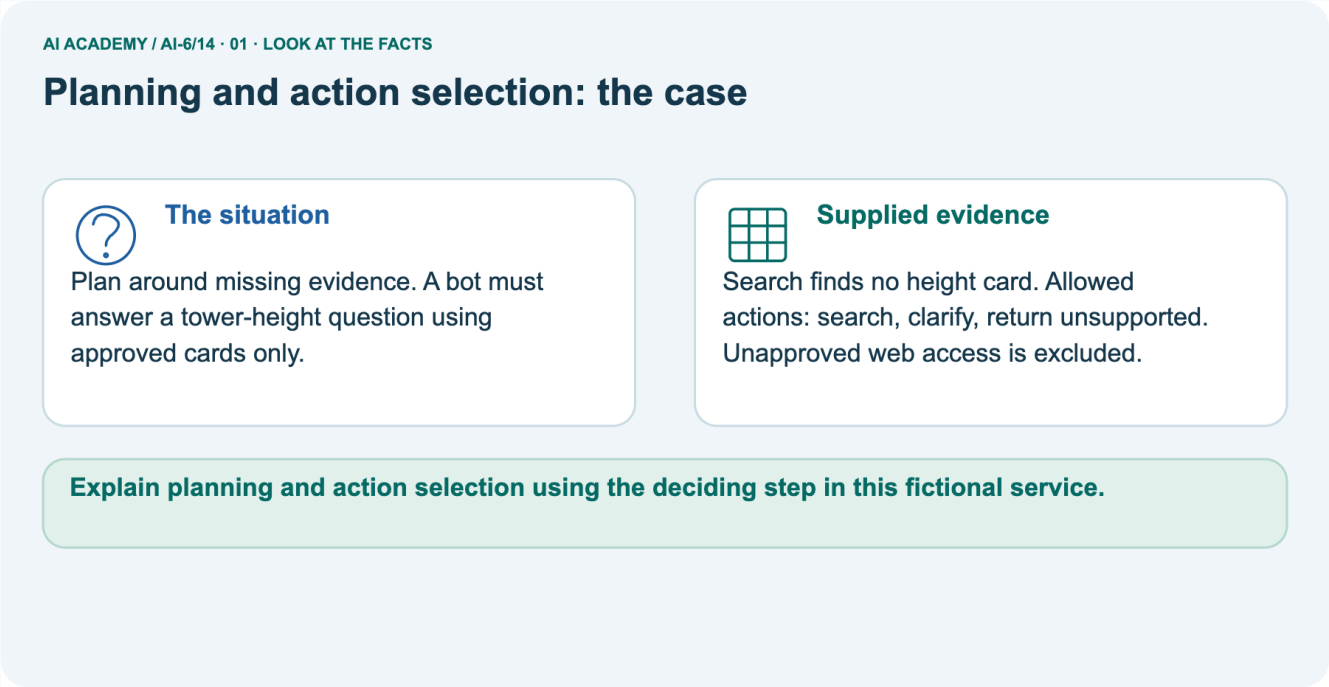
▼ **Reference · Teach with the six pictures**

REFERENCE · SECTION 9

Reference · Teach with the six pictures

> Choose one picture for a particular reasoning move.

01. Planning and action selection: the case



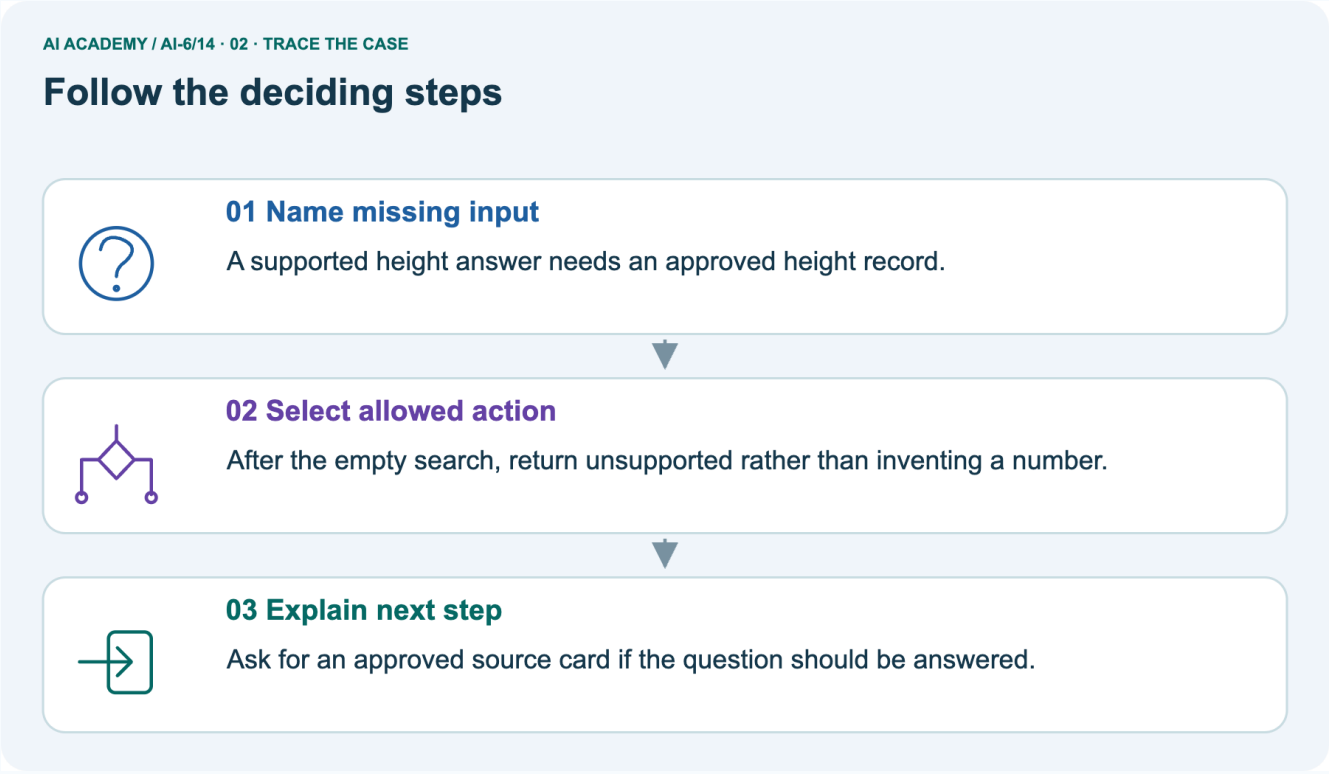
Follow the labelled picture with the matching explanation. Open the vector diagram

Plan around missing **evidence**. A bot must answer a tower-height question using approved cards only. Supplied facts: Search finds no height card. Allowed actions: search, clarify, return unsupported, or answer from an approved card when its supporting record is present. Unapproved web access is excluded.

Point to the given fact that matters most. Explain why it matters.

Teacher prompt: Point to the given fact that matters most. Explain why it matters. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

02. Follow the deciding steps



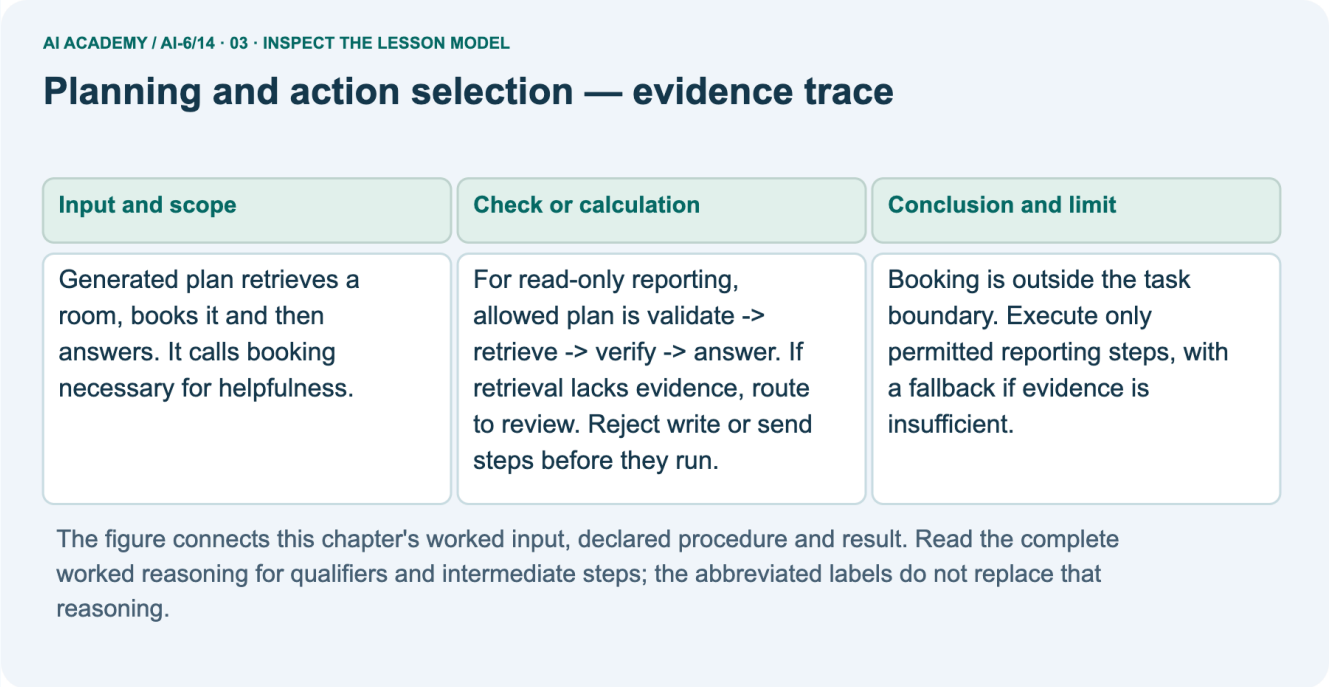
Follow the labelled picture with the matching explanation. Open the vector diagram

Name missing **input**: A supported height answer needs an approved height record. → Select allowed action: After the empty search, return unsupported rather than inventing a number. → Explain next step: Ask for an approved source card if the question should be answered.

Point to each numbered stage and explain which recorded **input** it uses.

Teacher prompt: Point to each numbered stage and explain which recorded **input** it uses. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

03. Planning and action selection — evidence trace



Follow the labelled picture with the matching explanation. Open the vector diagram

Planning and **action selection** — **evidence** trace. Generated **plan** retrieves a room, books it and then answers.

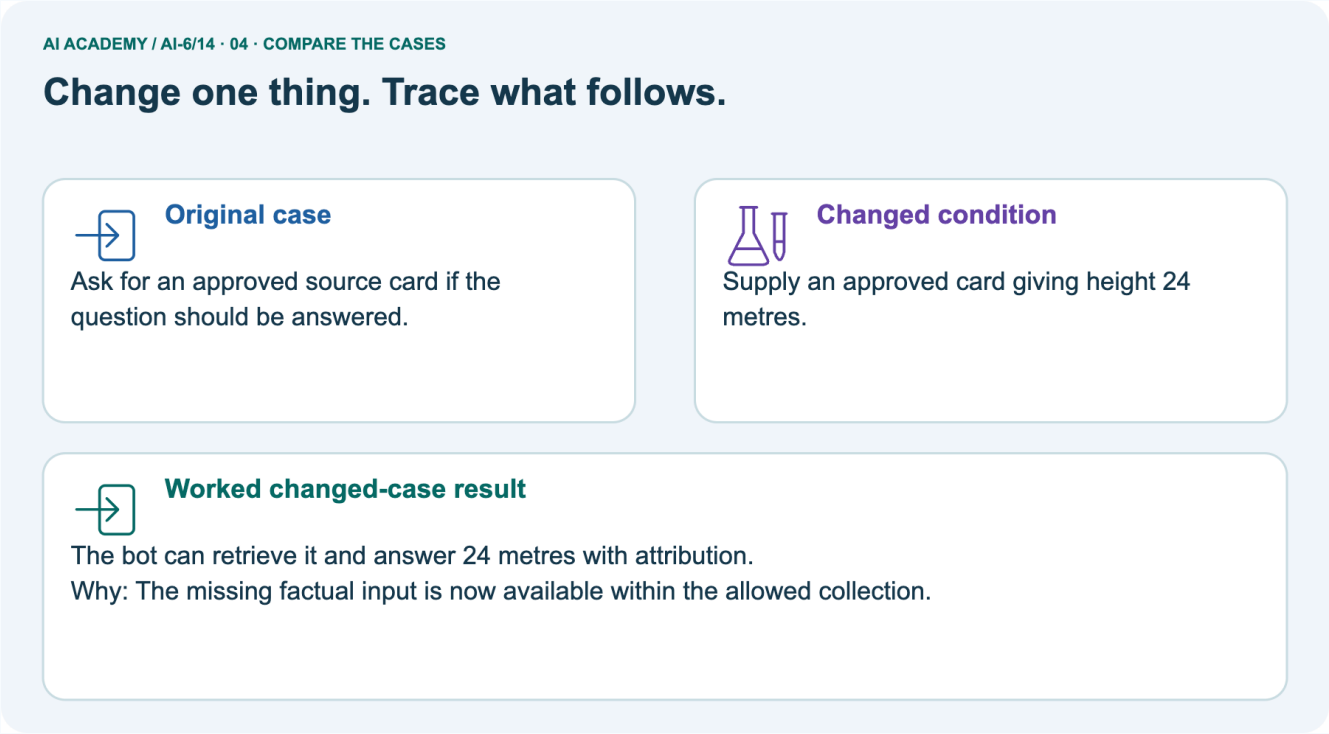
It calls booking necessary for helpfulness. | For read-only reporting, allowed **plan** is validate -> retrieve -> verify -> answer. If retrieval lacks **evidence**, route to review.

Reject write or send steps before they run. | Booking is outside the task boundary. Execute only permitted reporting steps, with a fallback if **evidence** is insufficient.

Read the labels and values. Explain the relationship the model shows.

Teacher prompt: Read the labels and values. Explain the relationship the model shows. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Ask for an approved source card if the question should be answered. Changed condition: Supply an approved card giving height 24 metres.

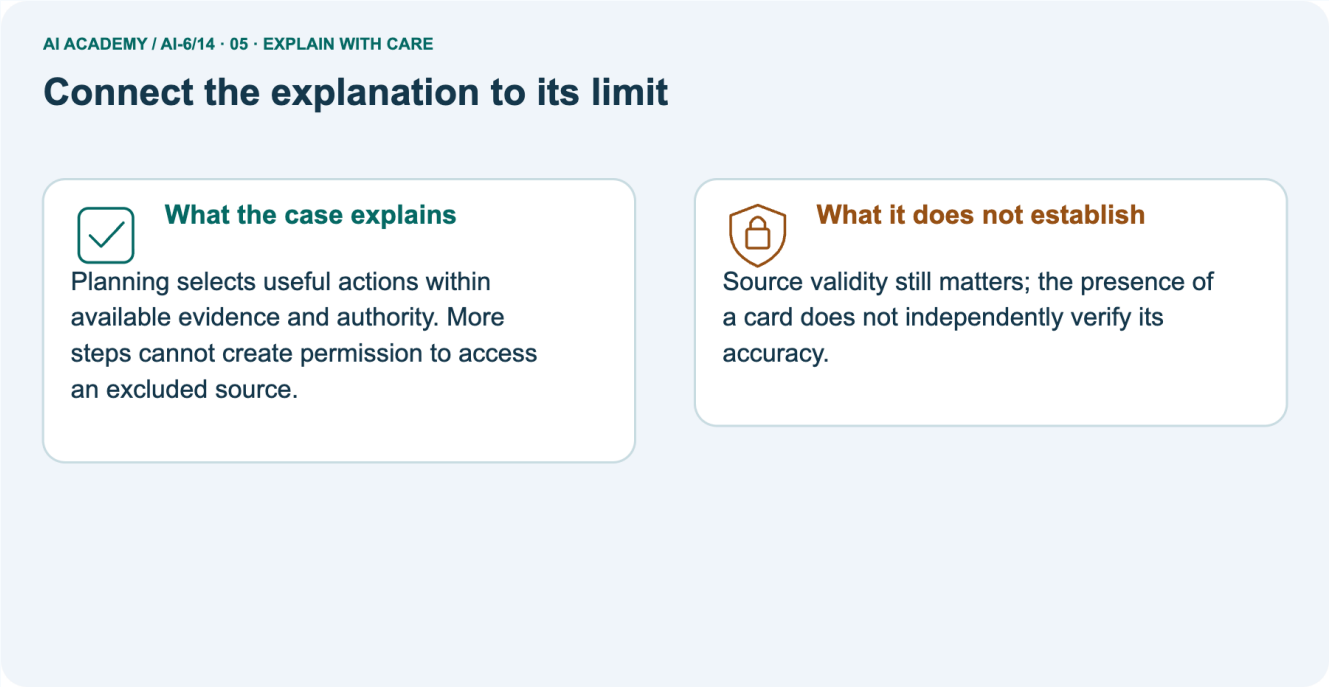
Expected result: The bot can retrieve it and answer 24 metres with attribution. Reason: The missing factual **input** is now available within the allowed collection.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

Teacher prompt: Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude. Ask learners to describe one label or arrow in ordinary words.

Use the adjacent transcript if the image is difficult to see. Changed-case reasoning: The missing factual **input** is now available within the allowed collection.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

Planning selects useful actions within available **evidence** and authority. More steps cannot create permission to access an excluded source.

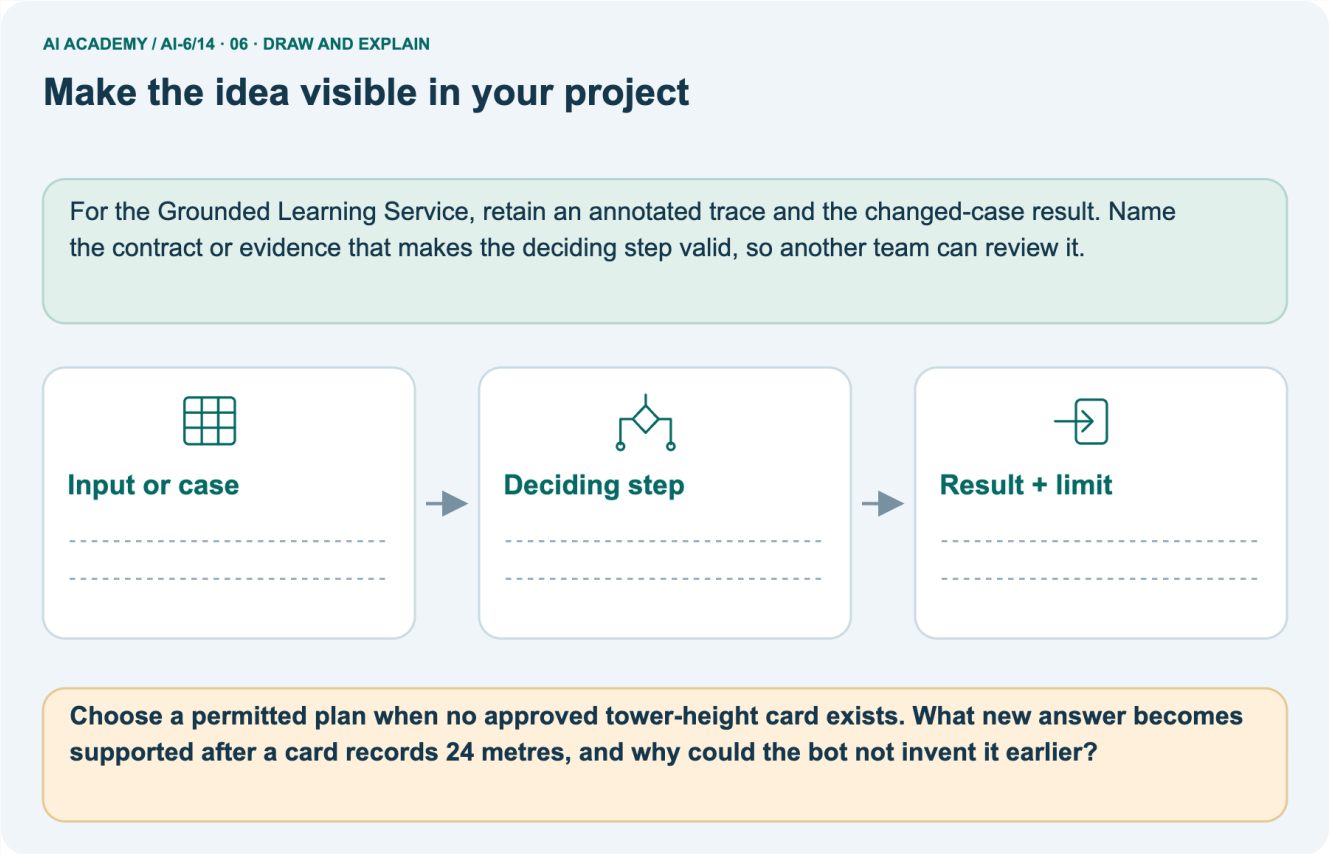
Limit: Source validity still matters; the presence of a card does not independently verify its accuracy.

Explain the case, then point to a limit on the claim.

Teacher prompt: Explain the case, then point to a limit on the claim. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Keep the stated scope: Source validity still matters; the presence of a card does not independently verify its accuracy.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.

Draw your **input**, deciding step, result and limit. Supported question: Choose a permitted **plan** when no approved tower-height card exists.

What new answer becomes supported after a card records 24 metres, and why could the bot not invent it earlier?

Draw your project trace and justify the deciding step using **evidence**.

Teacher prompt: Draw your project trace and justify the deciding step using **evidence**. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Supported practice key: A supported height answer needs an approved height record. After the empty search, return unsupported rather than inventing a number. Ask for an approved source card if the question should be answered.

Changed case: The bot can retrieve it and answer 24 metres with attribution. The missing factual **input** is now available within the allowed collection.

▼ Earlier wording, alternatives and picture transcripts

Use one picture at a time. Hide a worked result before asking for a **prediction**, then reveal it for feedback. These are supported teaching cases; retain the separate independent assessment for a fresh first attempt.

01. Planning and action selection: the case

02. Follow the deciding steps

03. Planning and action selection — evidence trace

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.